Practice Set A

Class 12 Mathematics, Chapter 3: Matrices

Maximum marks: **49** **20** questions

Name _____ Date _____ Score _____ / 49

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE SQP 2025-26 — 1 mark

If for three matrices $A = [a_{ij}]_{m\times 4}$, $B = [b_{ij}]_{n\times 3}$ and $C = [c_{ij}]_{p\times q}$, the products AB and AC are both defined and are square matrices of the same order, then the values of m, n, p and q are:

(A) $m = q = 3$ and $n = p = 4$ **(B)** $m = 2,\ q = 3$ and $n = p = 4$

(C) $m = q = 4$ and $n = p = 3$ **(D)** $m = 4,\ p = 2$ and $n = q = 3$

2 MCQ, CBSE 2023 — 1 mark

If $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$ and $A = B^2$, then x equals:

(A) ± 1 **(B)** -1

(C) 1 **(D)** 2

3 MCQ, CBSE SQP 2024-25 — 1 mark

If $A = \begin{bmatrix} 0 & 1 & c \\ -1 & a & -b \\ 2 & 3 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the value of $a + b + c$ is:

(A) 1 **(B)** 2

(C) 3 **(D)** 4

4 MCQ, CBSE 2023 — 1 mark

If $A = [a_{ij}]$ is a square matrix of order 2 such that $a_{ij} = 1$ when $i \neq j$ and $a_{ij} = 0$ when $i = j$, then A^2 is:

(A) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q No.	Question	Marks
5	MCQ, CBSE 2023 If for a square matrix A, $A^2 - A + I = O$, then A^{-1} equals: **(A)** A **(B)** $A + I$ **(C)** $I - A$ **(D)** $A - I$	1 mark
6	Assertion-Reason, Board-pattern Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. **Assertion (A):** $(A+B)^2 = A^2 + B^2$. **Reason (R):** For these matrices A and B, $AB = BA$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find the values of x and y if $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$.	2 marks

Level 2 — 18 marks

8	Short answer, Board-pattern Construct a 3×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \lvert 2i - 3j \rvert$.	2 marks
9	Short answer, PYQ-style If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1}.	3 marks
10	Short answer, PYQ-style Express the matrix $A = \begin{bmatrix} 2 & 0 \\ 5 & 4 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.	3 marks
11	Short answer, PYQ-style Find a matrix A such that $2A - 3B + 5C = O$, where $B = \begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$.	2 marks
12	Short answer, PYQ-style Find x, if $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = O$.	3 marks
13	Short answer, Board-pattern If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, find α $\left(0 < \alpha < \frac{\pi}{2}\right)$ such that $A^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.	2 marks

Q No.	Question	Marks
14	Short answer, Board-pattern If $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$.	3 marks

Level 3 — 23 marks

15	Long answer, PYQ-style If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, find the value of k such that $A^2 = kA - 2I$.	3 marks
16	Long answer, Board-pattern A factory makes three items P, Q and R. One unit of P needs 2 kg of raw material and 3 hours of labour; Q needs 1 kg and 4 hours; R needs 3 kg and 2 hours. Raw material costs ₹50 per kg and labour costs ₹40 per hour. Using matrix multiplication, find the total cost of making 20 units of P, 30 units of Q and 40 units of R.	3 marks
17	Long answer, PYQ-style If $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$, show that $A^2 = A$. Hence find $(A + I)^2$.	3 marks
18	Long answer, PYQ-style If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, show that $F(x)\,F(y) = F(x+y)$. Hence show that $[F(x)]^{-1} = F(-x)$.	5 marks
19	Long answer, PYQ-style If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, verify that $A^3 - 6A^2 + 9A - 4I = O$. Hence find A^{-1}.	5 marks
20	Case study, Board-pattern **Case Study:** A company makes three products x, y and z and sells them in two markets. In a year, Market I buys 10,000 units of x, 2,000 units of y and 18,000 units of z. Market II buys 6,000 units of x, 20,000 units of y and 8,000 units of z. The selling prices are ₹2.50, ₹1.50 and ₹1.00 per unit of x, y and z respectively.	4 marks

(i) Write the yearly sales as a 2×3 matrix and the selling prices as a column matrix. — 1 mark

(ii) Find the revenue from Market I. — 1 mark

(iii)(a) Using matrix multiplication, find the revenue from each market and the total revenue from both markets. — 2 marks

OR

(iii)(b) If the costs of making one unit of x, y and z are ₹2.00, ₹1.00 and 50 paise respectively, find the gross profit in each market using matrix algebra. — 2 marks

Answer key

Chapter 3: Matrices, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1 (A) $m = q = 3$ and $n = p = 4$

2 (C) 1

3 (A) 1

4 (D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

5 (C) $I - A$

6 (C) Assertion (A) is true, but Reason (R) is false.

7 $x = 3, y = -4$

Level 2

8 $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \\ 3 & 0 \end{bmatrix}$

9 $A^2 - 5A + 7I = O$; $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

10 $A = \begin{bmatrix} 2 & \frac{5}{2} \\ \frac{5}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$

11 $A = \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix}$

12 $x = -2$ or $x = -14$

13 $\alpha = \frac{\pi}{4}$

14 $(AB)' = B'A' = \begin{bmatrix} 5 & -7 \\ 8 & -1 \end{bmatrix}$, hence verified.

Level 3

15 $k = 1$

16 Total cost $=$ ₹19,900

17 $A^2 = A$; $(A + I)^2 = 3A + I = \begin{bmatrix} 7 & -6 & -12 \\ -3 & 10 & 12 \\ 3 & -6 & -8 \end{bmatrix}$

18 $F(x)F(y) = F(x + y)$ and $[F(x)]^{-1} = F(-x)$

19 $A^3 - 6A^2 + 9A - 4I = O$; $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$

20 Revenue: Market I ₹46,000, Market II ₹53,000, total ₹99,000. Gross profit: Market I ₹15,000, Market II ₹17,000.